Linear Regression

Unit 3: Fitting It, Reading It, and Answering the Question You Were Asked

Stat 220 | Statistics for Data Science

Where we left off

- Unit 1 compared two groups and asked whether the gap was bigger than the wobble.
- Now the question changes shape. Instead of two buckets we have a *relationship*: as one number moves, what happens to another?
- The machinery does not change. We will still produce an estimate, a standard error, and an interval, and we will still ask what chance alone would produce.

Principle: same template, new estimate

A slope is an estimate like any other. Everything you learned about reading a difference applies to reading a slope.

Why regression matters

- It does **two jobs at once**: it *predicts* ($\hat{y}$ for a new case) and it *explains* (how y relates to each x).
- Almost every model you'll meet, from a simple line to a deep network, is some descendant of "draw the best relationship between inputs and an output."
- And the interesting questions aren't about the formula. They're about which variables belong in the model, and what a coefficient really means once they're there.

In practice: the shape of this unit

"You added a variable and another coefficient flipped sign. Which number do you believe, and why?"

What this unit gives you

Things to know

- what a regression line is, and what least squares actually minimizes
- how to read a slope and an intercept *with their units*
- that a slope is an estimate, so it carries a standard error
- how binary and categorical predictors enter a model
- what an interaction means, in words
- what happens to a coefficient when you add or remove another predictor

Mistakes to stop making

- reading a slope without checking its units or its range
- treating a slope as the effect of intervening
- confusing a significant slope with a useful model
- chasing R^2 as if it measured correctness
- comparing coefficients on variables measured in different units
- controlling for a variable without asking what it does in the causal story

Regression to the mean

Principle: selecting on an extreme score

When you pick a group out because it scored at one end, some of what put it there was skill and some was luck. Skill shows up again next time. Luck does not. So the group drifts back toward the middle on its own, with nothing done to it and nobody helping.

Principle: a predictor can be real and still be circumstantial

Suppose the low scorers get an intervention and improve. "Received the intervention" would sit in a regression with a positive coefficient and a small p-value, and it would still not be doing anything. It predicts because of *how the group was chosen*, not because of what was done to them.

Where the word "regression" comes from

- Francis Galton, 1886, measuring the heights of parents and their grown children.
- Tall parents had tall children, but on average **closer to the middle** than the parents were. Same for short parents in the other direction.
- He called it *regression toward mediocrity*, and that phrase is why the technique this entire unit is about is called **regression**.

Principle: what Galton nearly got wrong

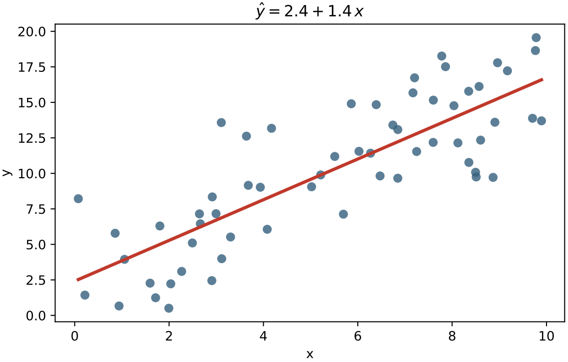
He first read it as a force pulling humanity toward the average. It is not a force. It is what happens to any measurement that is part skill and part luck, which is nearly all of them.

The arc of an analysis

Almost every regression analysis follows the same order, and the order matters more than the syntax:

1. **Pin down the question.** Predict, or explain? Whose effect do you actually care about?
2. **Know the data.** What is each variable, in what units, and how was it collected (assigned in an experiment, or just observed)?
3. **Look first.** Plot it. Check the range, the outliers, and any obvious lurking variable.
4. **Build the model the question needs.** Which predictors belong, and why each one?
5. **Check it.** Residuals, and the assumptions from Part 1.
6. **Read the answer.** The coefficient as a sentence, its uncertainty, and the caveats: confounding, extrapolation, causal or not.

What regression actually is



We draw the line that fits the cloud best:

$$\hat{y} \;=\; b_0 + b_1\,x.$$

- b_0 (intercept): predicted y when $x = 0$.
- b_1 (slope): how much $\hat{y}$ moves when x goes up by one unit.

That single number b_1 is usually the thing you actually care about.

Reading the slope and intercept

- A slope always carries units: "*units of* y per one *unit of* x."
- "$b_1 = 1.7$" is meaningless until you say "1.7 *inches of height* per one *shoe size*."
- The intercept is only meaningful if $x = 0$ is meaningful. A predicted height at shoe size 0 is nonsense, and that is fine. The intercept only positions the line.

Principle: interpretation is a sentence, not a number

Always state a slope as a full sentence with units and direction: "each extra X is associated with b_1 more Y." If you can't say the sentence, you don't understand the model yet.

Reading a fitted line out loud

Suppose a fit on housing data gives

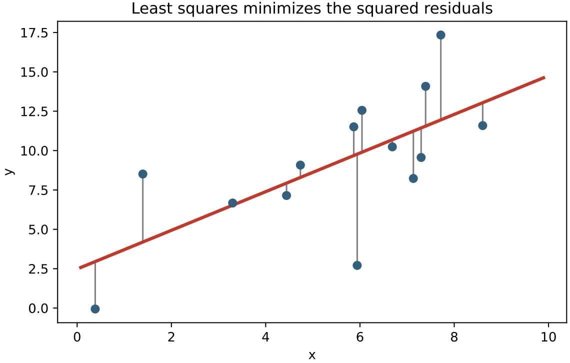
$$\widehat{\text{price}} = 41{,}000 + 118 \times \text{square feet.}$$

- **Slope:** each extra square foot is associated with \$118 more in price, *among houses like these*. Units: dollars per square foot.
- **Intercept:** the fitted price of a zero-square-foot house. Not a house. Intercepts are often meaningless alone. They position the line.
- **Range matters:** if the data runs 800 to 3,000 square feet, the line says nothing about a 12,000-square-foot mansion.

Trap: the units trap

"The coefficient is 118, so it is a big effect." Big compared to what? Switch to square *meters* and the same relationship reports 1,270.

Residuals and least squares



- A **residual** is the vertical gap between a point and the line: what the model missed for that case.
- "Least squares" picks the line that makes the sum of *squared* residuals as small as possible.
- Squaring is why a few far-off points (outliers) can pull the line toward them.

How much did the line actually explain?

Before the line, your best guess for any case was $\bar{y}$, and the spread around it is the **total variation**. After the line, what is left over is the residuals.

$$R^2 = 1 - \frac{\text{variation left in the residuals}}{\text{total variation in } y}$$

- $R^2 = 0$: the line did no better than just guessing $\bar{y}$ every time.
- $R^2 = 1$: every point sits exactly on the line.
- With one predictor, R^2 is the square of the **correlation** r, which runs from -1 to $+1$ and keeps the direction that squaring throws away.

Principle: three questions, three numbers

The **slope** says how much y moves. The p-**value** says whether the slope could be nothing. R^2 says how tightly the points hug the line. Knowing one tells you almost nothing about the other two.

What a slope answers

- "Does price go up with size?" The slope answers it, with a sign and a magnitude.
- "How much more do customers spend per extra email?" The slope answers it.
- Prediction uses the whole line. Explanation usually lives in a single slope.

Principle: keep the question in view

Before you read any coefficient, say the question it answers out loud. The math is the easy part. Knowing which slope answers your question is the skill.

Your turn #1

Your turn: say the sentence

A coffee chain fits

$$\widehat{\text{daily sales}} = 480 + 32\,(\text{loyalty emails sent}).$$

Turn to a neighbor and practice explaining, in one clear sentence with units, what the **32** means. Then: would you keep sending emails forever?

Your turn #1: answer

- "Each additional loyalty email is associated with about **$32 more** in predicted daily sales." Units and direction, in a sentence.
- "Forever?" No. The data only covers the range we actually observed. Pushing x far past that is extrapolation, and the line has no idea what happens out there.

Principle: a slope is local

A linear fit describes the relationship *inside the range of your data*. Read it there, not in a fantasy world far outside it.

The slope is an estimate, so it has a standard error

- Run the study again with new data and you'd get a slightly different slope. So b_1 is a random quantity with a sampling distribution, just like a mean.
- Its spread is $\widehat{\mathrm{SE}}(b_1)$, and the familiar template comes right back:

$$t = \frac{b_1 - 0}{\widehat{\mathrm{SE}}(b_1)}.$$

Principle: same engine, new question

Testing a slope is the Coke test again: assume the slope is really 0 ("x doesn't matter"), then ask how surprising a slope this big would be.

Testing whether the slope is zero

- **Null:** $b_1 = 0$, meaning x has no linear relationship with y.
- The software hands you b_1, its SE, the t, and a p-value. The p-value means exactly what it did in Part 1: how surprising this slope would be if the true slope were 0.
- For our scatter: $b_1 = 1.43$, $\widehat{\text{SE}} = 0.13$, so $t \approx 11$ and p is tiny. The upward relationship is not a fluke.

In practice: say it cleanly

"A small p on the slope says the relationship is hard to explain by chance. It does not say the relationship is large or useful. That's a separate question."

The traps from Unit 1, again

- With enough rows, a slope of essentially zero can still be "highly significant." Significance is partly a statement about your sample size.
- A confidence interval for the slope is almost always the better thing to report: $b_1 \pm (\text{critical } t) \times \widehat{\text{SE}}(b_1)$.

Trap: R^2 is not significance, and significance is not importance

A slope can have $p < 0.001$ and an R^2 of 0.01: a real relationship that explains almost none of the variation. Significant and important are different words.

Scenario: significant slope, tiny R^2

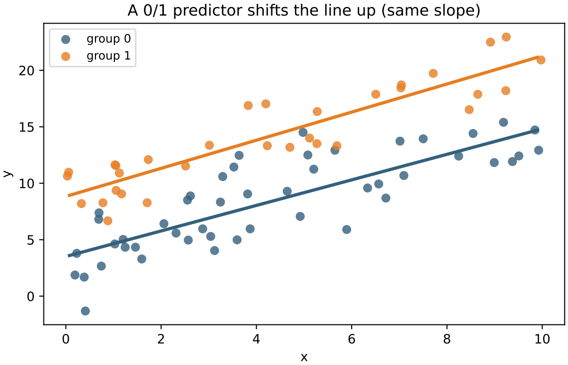
Setup. On 200,000 customers, "ads seen" predicts "spending" with $p < 10^{-9}$, slope $= \$0.02$ per ad, and $R^2 = 0.004$.

- The relationship is real (huge sample, tiny p) but explains almost nothing, and the per-ad effect is two cents.
- "Significant" here is mostly a story about having 200,000 rows.

In practice: how to answer

"It's a real but tiny relationship. I'd report the effect size and CI, note that R^2 is near zero so ads barely move spending, and let the business decide if two cents an ad is worth it."

A 0/1 predictor: the dummy variable



Code a yes/no variable as 1/0 and drop it in the model:

$$\hat{y} = b_0 + b_1 x + b_2\,(\text{group}).$$

- b_2 shifts the whole line up or down.
- You get two **parallel** lines, one per group.
- b_2 is the gap between them, holding x fixed.

The dummy coefficient is a group comparison

- b_2 answers: "at the same x, how much higher is group 1 than group 0?"
- With no x in the model at all, that coefficient is just the difference in group means. It is the two-sample comparison from Unit 1, written as a regression.

Principle: regression unifies the tests

A two-group t-test is a regression on a single 0/1 predictor. Same estimate, same SE, same p-value. Once you see this, the whole toolkit collapses into one idea.

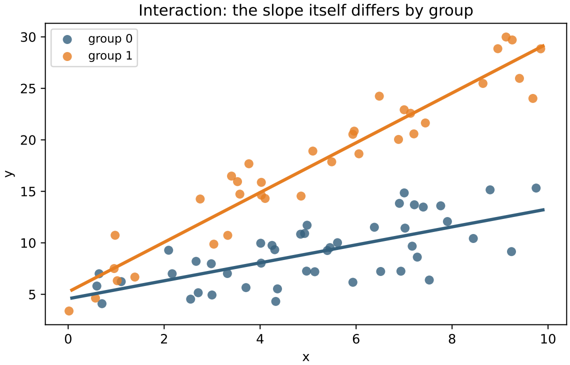
Categories with more than two levels

- For a variable like region (West, Mountain, Midwest, . . .), pick one level as the **baseline** and add one 0/1 dummy for each *other* level.
- Each coefficient then means "this level versus the baseline, holding everything else fixed."
- Four regions means three dummies. The baseline is folded into the intercept.

Trap: baseline confusion

Every categorical coefficient is relative to the baseline, not to zero and not to the other levels. If you forget which level is the baseline, every interpretation is wrong.

When the slope itself differs by group: interaction



A dummy shifts the line. An **interaction** (a group $\times$ x term) lets the *slope* change too.

- Lines are no longer parallel.
- Now "the effect of x" depends on the group.

We'll go deeper on interactions in Part 3.

The interaction model, written out

$$\hat{y} = b_0 + b_1 x + b_2\, G + b_3\, (G \times x), \qquad G = 0 \text{ or } 1$$

Put in each value of G and the two lines fall out:

Group	**Model becomes**	**Intercept**	**Slope**
$G = 0$	$b_0 + b_1 x$	b_0	b_1
$G = 1$	$(b_0 + b_2) + (b_1 + b_3)\, x$	$b_0 + b_2$	$b_1 + b_3$

- b_2 is the gap in *intercept*. b_3 is the gap in *slope*.
- Testing $b_3 = 0$ is exactly the question "does the effect of x depend on the group?"

Trap: b_1 quietly changes meaning

With the interaction in the model, b_1 is no longer "the effect of x." It is the effect of x **in the group coded 0**. Quoting it as the overall slope is the usual way this model gets misread.

Your turn #2

Your turn: interpret the coefficient

A model of monthly rent gives

$$\widehat{\text{rent}} = 600 + 0.9\,(\text{sqft}) + 250\,(\text{has parking}),$$

where *has parking* is 1 if yes, 0 if no. Practice explaining to a neighbor what the **250** means. Be careful to say what is being held fixed.

Your turn #2: answer

- "Among units *of the same size*, having parking is associated with about **$250 more** rent per month."
- The phrase "of the same size" is the whole point: the coefficient holds square footage fixed. Drop that phrase and you've changed the claim.

Principle: coefficients are conditional

Every coefficient in a multi-predictor model means "holding the other predictors fixed." Saying that phrase out loud is what separates a correct interpretation from a wrong one.

Multiple regression, quickly

With more than one predictor,

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k,$$

- each coefficient is the effect of *its* predictor **holding all the others fixed**.
- That little phrase is where all the statistical thinking lives. The same slope can look completely different depending on what else is in the model.

Principle: marginal vs. partial

A slope with x alone (marginal) and the same slope with other variables added (partial) are answers to *different questions*. Neither is "the truth" on its own.

Confounding

- A **confounder** is a third variable that drives both x and y.
- It can manufacture a relationship that isn't really about x at all, or hide one that is.
- Example: ice cream sales predict drowning deaths. Not because ice cream is dangerous, but because *summer* drives both.

Principle: the question to always ask

Before trusting a slope as "the effect of x," ask: what else moves with x that also moves y? That third variable is where slopes go to die.

Your turn #3 (vote)

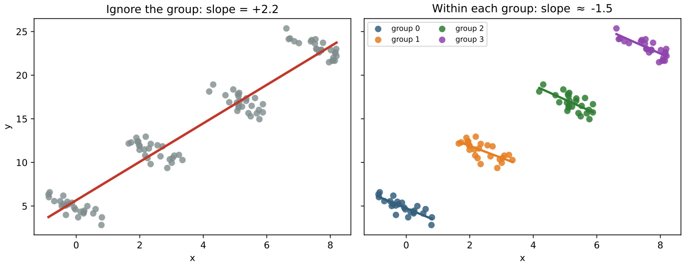
Your turn: predict the within-group slope

A health study finds that, across everyone, **more weekly exercise** goes with **higher cholesterol**: a clearly positive slope.

Now we split people by **age group** and look *within* each one. Vote:

(A) still positive **(B)** flat **(C)** flips negative

Your turn #3: answer



Ignore age: slope is **positive** ($\approx +2.2$).

Within each age group: slope is **negative** (≈ -1.5). Exercise lowers cholesterol, but older people both exercise more and run higher.

This sign flip is **Simpson's paradox**. Age was the confounder all along.

Adding a predictor

- Adding a **confounder** (like age above) removes bias. The slope you get after controlling for it answers a cleaner question.
- But adding the *wrong* variable can create bias: a **mediator** (a step on the path from x to y) or a **collider** (something x and y both cause).

Trap: "control for everything" is not a strategy

Throwing every available variable into the model can fix confounding or cause new bias, depending on each variable's role. You need a reason for each one, not a reflex.

Removing a predictor: omitted-variable bias

- Leave a real confounder out and its influence doesn't vanish. It gets absorbed into the slopes of whatever predictors remain.
- That is **omitted-variable bias**: the slope you report is contaminated by the variable you didn't include.
- This is why coefficients move, sometimes dramatically, when you add or drop a single variable.

In practice: a classic

"A coefficient flipped sign when I added one variable. Which is right?" Answer: "Whichever model controls for the actual confounders. The sign flip is the model telling you a confounder was hiding in the first version."

A coefficient that flips

The workflow. You are modeling fuel economy. You start with how long a car takes to reach 60 mph.

Model	acceleration	horsepower	R^2
`mpg ~ acceleration`	$+\mathbf{1.198}$ ($p = 2 \times 10^{-18}$)		0.18
`mpg ~ acceleration + horsepower`	$-\mathbf{0.610}$ ($p = 6 \times 10^{-7}$)	-0.188	0.63

- Alone, slower cars get *better* mileage: $+1.2$ mpg per extra second. Sensible.
- Then a colleague says you have to control for engine power. Obviously right.
- You add horsepower, and acceleration **flips to** -0.61.

Both coefficients are overwhelmingly significant. Neither model is broken, and this is not noise.

Why it flipped

Leave out a variable and its influence does not disappear. It rides in on whatever you kept:

$$\underbrace{b_{\text{acc}}^{\text{alone}}}_{+1.198} = \underbrace{b_{\text{acc}}^{\text{full}}}_{-0.610} + \underbrace{b_{\text{hp}}}_{-0.188} \times \underbrace{\delta}_{-9.616}$$

- δ is the omitted variable regressed on the one you kept: each extra second to 60 mph goes with 9.6 fewer horsepower. Each horsepower costs 0.188 mpg.
- Two negatives multiply to a bias of $+\mathbf{1.807}$, three times the real -0.610, so it does not shrink the coefficient, it overturns it. Drop horsepower and it returns.

Principle: both are right, about different questions

Across *all* cars, do slower ones get better mileage? Yes, they have small engines. Among cars of the *same* horsepower? No: still sluggish at fixed power means heavy, and weight burns fuel.

Correlation is not causation, stated precisely

- A regression slope is a **description**: how y moves with x in this data.
- It becomes a **causal** statement only under an assumption you bring from outside: that there are no lurking confounders.
- Randomized experiments (like our Coke test) earn that assumption by design. Observational data does not.

Principle: where causation comes from

Causation is not something a slope proves. It comes from the *design* of how the data was collected, or from assumptions you state and defend.

Scenario: "does X cause Y?"

Setup. A dashboard shows: users who get the onboarding email spend 20% more. A PM says "the email causes more spending, send it to everyone."

- Who *gets* the email may already differ: more engaged users may opt in, and engaged users spend more anyway. That's a confounder.
- The clean answer is an experiment: randomly send the email to half.

You may be asked: how to answer

"The 20% is a correlation that could be confounded by who opts in. Before claiming the email causes spending, I'd run a randomized test, or at least control for engagement and check whether the gap survives."

From question to answer: a worked example

Problem. A landlord asks: "does adding parking really raise the rent I can charge, or do parking units just happen to be bigger?" Walk the six steps:

1. **Question:** the *effect* of parking on rent (explanation), not prediction.
2. **Data:** rent, square footage, parking (0/1), one row per unit, all observed (nothing was randomly assigned).
3. **Look:** parking units do tend to be larger, so size is a likely confounder.
4. **Model:** regress rent on parking *and* size, so parking is read at a fixed size.
5. **Check:** residuals look fine, and no single building drives the result.
6. **Answer:** "about +\$250 holding size fixed," with its CI, plus the caveat that this is observational, so an unmeasured confounder could still remain.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM to interpret a regression

We told it: "Here is a regression of salary on years of education, from survey data. Interpret it."

It answered: "Each extra year of education causes about $4,200 more salary, so requiring more schooling would raise incomes."

Mistake, or no mistake? Decide before we click.

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Trap: verdict: mistake

Survey data is observational, so "causes" is unearned. Ability, family background, and field all confound education and salary, and the policy claim rides on that bad causal leap.

LLM check: mistake or not? (2 of 2)

LLM check: same idea, a different answer

We told it: "Interpret the coefficient on bedrooms in a house-price model that also includes square footage."

It answered: "Holding square footage fixed, each extra bedroom is associated with about $9,000 higher predicted price. This is associational, not necessarily causal."

Mistake, or no mistake?

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Principle: verdict: no mistake

This one is right: it holds the other predictor fixed, says "associated" instead of "causes," and flags the causal caveat. Don't assume the model is always wrong.

Fitting and reading a regression

1. **Plot y against x first.** A line on a curved cloud is a wrong answer with a good p-value.
2. **Fit it:** `sm.OLS(y, sm.add_constant(X)).fit()`, then read `.summary()`.
3. **Read the slope with its units**, and rescale to something a human feels (per 1,000 lbs, not per lb).
4. **Check the slope against its SE**, which is the same estimate-over-noise ratio as always.
5. **Get the interval**, `.conf_int()`, and report it rather than the p-value alone.
6. **Plot residuals against fitted values** to check the straight-line and constant-spread assumptions.
7. **State the range of x the data covers**, and refuse to predict outside it.

The traps, all in one place

Trap: the ones that bite most often

- Reading a slope with no units, no direction, no uncertainty.
- Treating R^2 as the score: high means good, low means useless.
- Confusing a significant slope with an important one.
- Interpreting a coefficient without saying “holding the others fixed.”
- Forgetting which level is the categorical baseline.
- Calling a slope “the effect” while a confounder is lurking.
- Controlling for everything, including mediators and colliders, on reflex.
- Reading an observational slope as proof of causation.

How to read any regression coefficient

Principle: the reusable structure

1. **Say the sentence:** "each extra (unit of x) is associated with b more (units of y), holding the other predictors fixed."
2. **Add the uncertainty:** the SE and the confidence interval.
3. **Separate** significance, size, and R^2. They answer different questions.
4. **Ask what's missing:** which confounder could be moving this slope?
5. **Decide:** description or causal claim, and is the design strong enough to say so?

That structure works for a coffee chain, a rent model, or a neural network's last layer.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Simple and multiple regression	`sm.OLS(y, sm.add_constant(X)).fit()`
Formula interface with categories	`smf.ols(ÿ ~ x + C(group); data)`
An interaction	`smf.ols(ÿ ~ x * group; data)`
Coefficient intervals	`fit.conf_int()`
Residual diagnostics	`fit.resid` against `fit.fittedvalues`
Correlation between predictors	`df[cols].corr()`
Transform a skewed variable	`np.log(y)`, then interpret on the log scale

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- We held the model fixed and read one slope at a time. **Part 3** is about *building* the model: several predictors at once, interactions, transformations, and the constant tension between fitting the data and overfitting it.
- The thread continues: every coefficient is still an estimate with uncertainty, and the hardest questions are still about which variables belong.

Principle: the through-line

A model is a set of answers, one per coefficient. Your job is to know which question each one answers, and which variable might be quietly changing the answer.